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Detecting subtle change from dense Landsat time series: Case studies of mountain pine beetle and spruce beetle disturbance

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ABSTRACT

In contrast to abrupt changes caused by land cover conversion, subtle changes driven by a shift in the condition, structure, or other biological attributes of land often lead to minimal and slower alterations of the terrestrial surface. Accurate mapping and monitoring of subtle change are crucial for an early warning of long-term gradual change that may eventually result in land cover conversion. Freely accessible moderate-resolution datasets such as the Landsat archive have great potential to characterize subtle change by capturing low-magnitude spectral changes in long-term observations. However, past studies have reported limited success in accurately extracting subtle changes from satellite-based time series analysis. In this study, we introduce a supervised framework named 'PIDS' to detect subtle forest disturbance from a comprehensive Landsat data archive by leveraging disturbance-based calibration sites. PIDS consists of four components: (1) Parameter optimization; (2) Index selection; (3) Dynamic stratified monitoring; and (4) Spatial consideration. PIDS was applied to map the early stage of bark beetle infestations (i.e., a lower per-pixel fraction of trees cover that show visual signs of infestation), which are a typical example of subtle change in conifer forests. Landsat Analysis Ready Data were used as the time series inputs for mapping mountain pine beetle and spruce beetle disturbance between 2001 and 2019 in Colorado, USA. PIDS-detection map assessment showed that the overall performance of PIDS (namely 'F1 score') was 0.86 for mountain pine beetle and 0.73 for spruce beetle, making a substantial improvement (> 0.3) compared to other approaches/products including Continuous monitoring of Land Disturbance, LandTrendr, and the National Land Cover Database forest disturbance product. A sub-pixel analysis of tree canopy mortality percentage was performed by linking classified high-resolution (0.3- and 1-m) aerial imagery and 30-m PIDS-detection maps. Results show that PIDS typically detects mountain pine beetle infestation when ≥56% of a Landsat pixel is occupied by red-stage canopy mortality (one year after initial infestation), and spruce beetle infestation when ≥55% is occupied by gray-stage mortality (two years after initial infestation). This study addresses an important methodological goal pertinent to the utility of event-based reference samples for detecting subtle forest change, which could be potentially applied to other types of subtle land change.

1. Introduction

In the last four decades, satellite remote sensing has become established as the key technology for providing synoptic and precise land change information (Kennedy et al., 2009; Rogan and Chen, 2004; Roy

et al., 2014). Methods for accurately detecting land-cover type conversion (e.g., deforestation), typically associated with high-magnitude spectral changes in satellite imagery, are well established and have been the focus of many important improvements in bi-temporal/multi-temporal change detection approaches (Coppin et al., 2004b; Hansen

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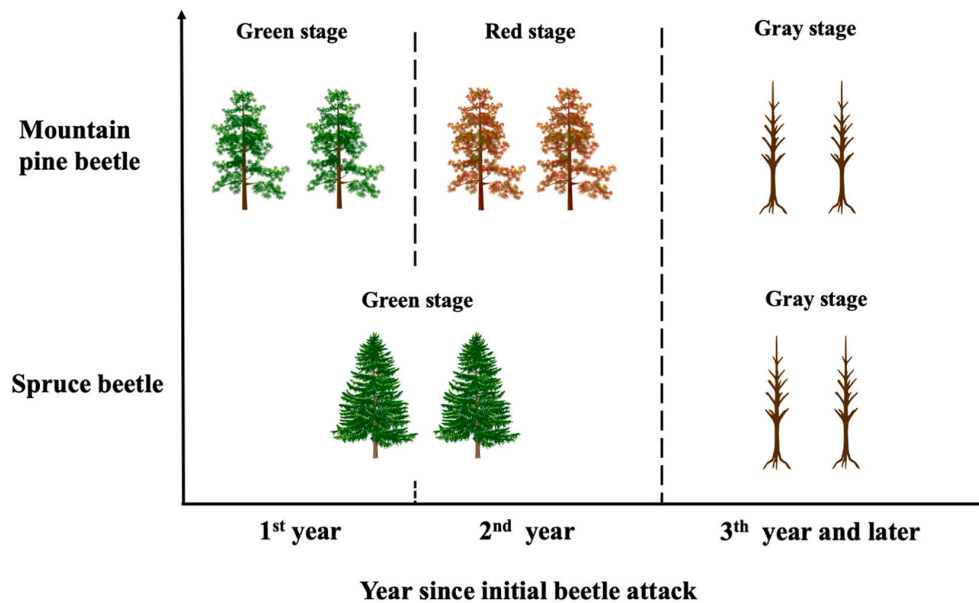


Fig. 1. Schematic diagram illustrating the stages of trees killed by mountain pine beetle and spruce beetle infestation.

and Loveland, 2012; Hussain et al., 2013; Zhu, 2017). In contrast, this paper focuses on detecting subtle changes in ecosystem dynamics, an emerging topic that is attracting a great deal of new attention in the remote sensing literature (Bullock et al., 2020b; Woodcock et al., 2020). Subtle changes, with regard to vegetation communities, are low-severity changes in the condition, composition, or structure within a persisting land-cover category (Coppin et al., 2004a; Woodcock et al., 2020). A variety of physical or biological processes lead to subtle changes on the terrestrial surface, especially at their initial stage, such as insect- or drought-induced forest condition change, grassland degradation, glacier melt, and soil moisture loss in cropland. Another commonly used term, 'gradual change,' defines a group of slow changes with a long duration of consequential impacts (Vogelmann et al., 2016; Vogelmann et al., 2012) such as climate-induced tree dieback (Allen et al., 2010; Yang et al., 2012). In contrast, 'subtle change' refers to a broader categorization for low-magnitude land changes that can be either short term (e.g., selective logging) or long term (e.g., forest succession) (Vogelmann et al., 2016). From an operational viewpoint, a confirmation of gradual change requires a collection of long time series data over several years to several decades, whereas the detection of subtle change only requires a short period of data collection to enable an alert for disturbance detection. Accurate and timely mapping of subtle change is important for early warning of the initiation stage for long-term gradual changes that may eventually result in dramatic land-cover transition. For example, persistent drought stress may lead to the conversion of forest to shrubland or grassland (Huang and Anderegg, 2012), and semi-arid grassland can be degraded to bare soil due to meteorological drought and grazing pressure (Liu et al., 2013).

Mountain pine beetle (MPB, *Dendroctonus ponderosae*) and spruce beetle (SB, *Dendroctonus rufipennis*) are two native bark beetles in North America that have caused extensive mortality in conifer forests across south-central Alaska (Sherriff et al., 2011), western Canada (Aukema et al., 2006) and the western United States (Hicke et al., 2020). The tree-level damage caused by bark beetles can be characterized by several infestation stages. The period immediately following MPB attack, but before the symptoms of infestation are visibly evident in the tree crown, is called the 'green stage' (Wulder et al., 2009). Despite lacking visible crown symptoms, green-stage trees experience increased water stress (Foster et al., 2017) because larvae feed on the phloem tissue and

introduce a fungus that disrupts the translocation of water and nutrients within the bole (Safranyik and Carroll, 2006). The 'red stage' follows the green stage as the green chlorophyll pigment molecules are significantly lost and foliage turns to red or yellow. Host trees ultimately drop all needles after 3–5 years, transitioning to the 'gray stage'. Different from MPB, spruce trees infested by SB do not exhibit a characteristic 'red stage' but change to green or yellow/green foliage (DeRose et al., 2011) for two years after initial infestation, and then drop their needles and branches (see Fig. 1). While beetle disturbance eventually leads to substantial multispectral change (usually visible in eyes) due to severe loss of overstory canopy, only minor and invisible spectral change of individual trees occurs at its green stage. Moreover, the stand-level beetle damage is often a continuous process that can occur over 3–7 years; when few killed trees exist within an image pixel, the reflectance of stressed foliage is mixed with other live foliage, branches, and soil background effects (Ahern, 1988). Hence, bark beetle infestations at its early stage (hereafter referred to none or a lower fraction of trees within a pixel that show visual signs of infestation) is a typical example of subtle changes (Kennedy et al., 2009; Meigs et al., 2011; Vogelmann et al., 2016).

It is critical for forest managers to acquire accurate location information on the early stage stands that can be used to prevent brood emergence and implement suppression strategies (Wulder et al., 2009). Previous efforts for detecting such subtle changes have used high spatial resolution multispectral (< 5 m) data (Coops et al., 2006; Immerzeel et al., 2014; Wiesmair et al., 2016) or hyperspectral data (Behmann et al., 2014; Niemann et al., 2015) using imagery captured prior to and following a known change event. High spatial resolution and hyperspectral data, however, are often impractical for a large-area analysis due to their limited geographic coverage, limited frequency, and high cost (Rogan and Chen, 2004; Ye et al., 2018). Furthermore, subtle changes can be obscured by noise from two different data acquisitions caused by differences in sun or view angle, phenological states, cloud cover and cloud shadow (Kennedy et al., 2009). Alternatively, benefiting from a short repeat acquisition cycle, extensive spatial coverage, and free acquisition costs, moderate resolution satellite imagery (5–30 m resolution) from Landsats 4–8 and Sentinel-2, have been recognized as important resources for subtle change detection and large-area monitoring through the identification of spectral departures in pixel-level

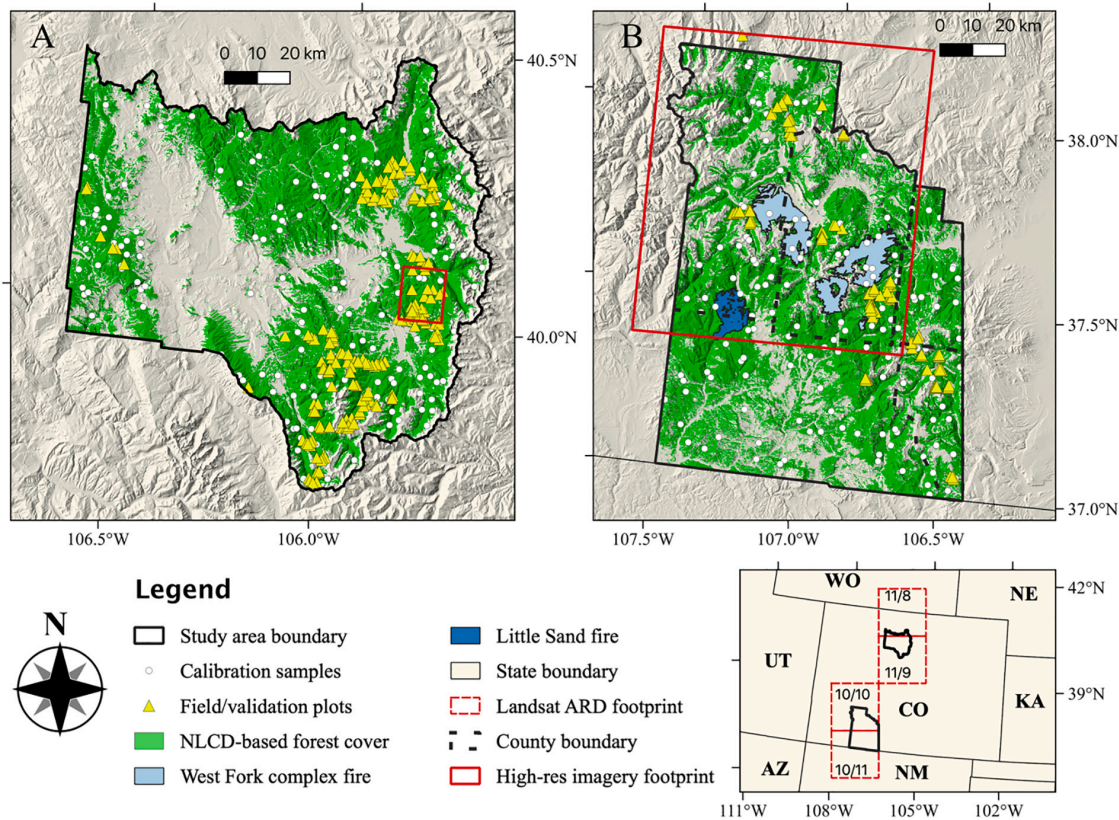


Fig. 2. Location of the study area sites in Colorado, that experienced mountain pine beetle (MPB; A) and spruce beetle (SB; B) infestation over the past 20 years. Each study site is covered by two Landsat Analysis Ready Data (ARD) tiles. The calibration (circles) and evaluation datasets (triangles) were generated and interpreted independently. Red rectangles indicate footprints of high-resolution aerial imagery used in this study. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

response history (Kennedy et al., 2014; Vogelmann et al., 2016; Woodcock et al., 2020). The Landsat data archive, known for its half-century of continuous data collection, provides an important baseline for subtle change detection. Compared to coarse-resolution datasets such as the Moderate Resolution Imaging Spectroradiometer (MODIS, 250–1000 m), the spatial resolution of the Landsat sensors is much finer (30 m), allowing for a larger proportion of pixels to represent pure land-cover types, which boosts the dynamic range of response and sensitivity to the timing of subtle change (Vogelmann et al., 2016).

A prerequisite for success in detecting subtle change is data consistency across different collection dates and sensors used in the analysis, because subtle-change signals are often weak and can be obscured by data noise (Vogelmann et al., 2016). Studies conducted to analyze the consistency among calibrated Landsat surface reflectance from Landsat-5 Thematic Mapper (TM), Landsat-7 Enhanced Thematic Mapper Plus (ETM+) and Landsat-8 Operational Land Imager (OLI) reported no notable inconsistency observed for both inter- and intra-sensor comparison (Li et al., 2013; Mishra et al., 2014; Qiu et al., 2019; Sulla-Menashe et al., 2016; Vogelmann et al., 2016), although the orbital drift issue of Landsat-5 TM might not be always negligible (particularly 1995–2000 and 2003–2007) (Zhang and Roy, 2016). Generally, the remarkable history of calibration activities makes an assemblage of a cross-sensor Landsat dataset readily available, supporting the generation of a dense time series for performing subtle change detection tasks (Vogelmann et al., 2016).

Continuous Change Detection and Classification (CCDC) (Zhu and Woodcock, 2014; Zhu et al., 2012) is an algorithm that was developed to detect different forms of land change from an assemblage of all available

Landsat 4–8 data. Recently, improvements have been made to CCDC to detect all types of land disturbances (Zhu et al., 2020), and to operate in near real-time scales (Ye et al., 2021). The dense trajectories (i.e., multiple data retrievals within a given year) of Landsat observations used by CCDC can illuminate the phenological responses at the pixel level, making it possible to capture subtle intra-annual change from the previous stable stage (Vogelmann et al., 2016). Another category of the common time-series approaches, such as LandTrendr (Kennedy et al., 2010) and Composite2Change (Coops et al., 2020), uses single-index, and annual Landsat time series, which have been applied to map subtle changes associated with insect infection (Coops et al., 2020; Meigs et al., 2011; Senf et al., 2015). These approaches require several consecutive years to confirm a temporal trend for subtle change events, and hence, are better suited for characterizing gradual and long-term change than subtle change.

To date, most recent CCDC-like algorithms have been developed to address subtle change detection issues and have shown substantially improvement compared to the original CCDC algorithm (Bullock et al., 2020b; Ye et al., 2021; Zhu et al., 2020). However, the reported omission errors are still relatively high for low-severity forest change. For example, Continuous monitoring of Land Disturbance (COLD) reported ~40% omission error rate for the ‘stress’ disturbance category based on forest inventory plots distributed across the conterminous United States (Zhu et al., 2020). The primary reason for high omission error is that these time-series algorithms apply a single set of model parameters (e.g., the change-magnitude threshold to CCDC-like algorithms) that mainly targets the detection of generic disturbance events. Explicit calibration of a time series-based algorithm, where the model parameters are

calibrated using a reference dataset for a specific disturbance type, could be critical in producing more accurate subtle change maps (Bullock et al., 2020a; Cohen et al., 2017). Another suggested improvement is optimizing the spectral band/index selection process because subtle-change signals are more separable using certain types of vegetation indices (VIs) (Cohen et al., 2020; Healey et al., 2005). Yet most current time series algorithms determine their spectral inputs either by prior knowledge (Verbesselt et al., 2010; Zhao et al., 2019) or by testing a limited number of VIs defined by the developers (Cohen et al., 2010; Zhu et al., 2020). Systematic approaches are still lacking for selecting the optimal group of spectral inputs in terms of detecting a specific change agent.

Beyond parameter calibration and index selection, it is essential to apply multiple model specifications (e.g., spectral variables and model parameters) stratified by thematic information (Vogelmann et al., 2016), because land-cover types differ in their specificity with respect to spectral transformation, having different sensitivities to change-magnitude thresholds (Rogan and Mietkiewicz, 2015). A subtle-change model is often only applied to certain land-cover types (e.g., forest), while others are better monitored using model specifications with less sensitivity to subtle change (e.g., grassland). This necessitates the dynamic updating of land cover in response to disturbance during the monitoring process, with the goal of determining when and where to apply specific models to monitor subtle change.

Two important questions remain unanswered: (1) how to improve detection of early beetle-infestation signals from Landsat time series (compared with other algorithms and datasets); and (2) what biological infestation stages are indicated by the early signals detected from Landsat time series. We addressed these critical knowledge gaps in two steps. First, a new methodological framework ('PIDS') was developed to more accurately map user-targeted subtle change from dense Landsat time series (Section 3.1). We evaluated the ability of PIDS to detect the 'earliest signals' associated with recent beetle disturbances in regions of Colorado, with a comparison to other existing time series approaches and disturbance products (Section 4.1).

Second, we explored what beetle infestation stages were present when the earliest signals were detected by PIDS (Section 4.2). High-resolution aerial imagery was leveraged to estimate the percentages of sub-pixel mortality present for different infestation stages when the signals were first detected by PIDS, considering the nature of mixed signals at a scale of 30-m (the method is detailed in Section 3.3).

2. Study area and data

2.1. Study area

Two sites were selected in Colorado, USA, one with a recent MPB disturbance (2002–2012) and the other with a recent SB disturbance (2004 – present). Grand County (4842 km², Fig. 2A) was chosen as the MPB site. The elevation for Grand County ranges from 2056 to 4151 m with an average of elevation of 2901 m. The forests in this area are predominantly even-aged stands of lodgepole pine and some spruce-fir forest stands, which are mainly located at elevations ranging from 2300 to 3300 m. An extensive and severe MPB outbreak began in Grand County, Colorado in 2002 (Tishmack et al., 2005). MPB-induced lodgepole pine mortality reached its peak between 2005 and 2008 in this region (Caldwell et al., 2013; Meddens and Hicke, 2014), and diminished by 2012 because of the depletion of host trees (Lester, 2020).

The SB site (10,241 km²) is in the spruce-fir zone of the eastern San Juan Mountains in southwestern Colorado (Fig. 2B), including Hinsdale County, Mineral County, Archuleta County, and parts of Rio Grande and Conejos Counties that were intersected by Landsat Analysis Ready Data (ARD) tiles 10/10 and 10/11. Elevation ranges between 1833 and 4372 m, with an average elevation of 3019 m. Forests are dominated by Engelmann spruce and interspersed with subalpine fir and quaking aspen (*Populus tremuloides*) (Andrus et al., 2020). The spruce beetle

outbreak initiated around 2004 and peaked between 2009 and 2013; almost 90% of spruce-fir forests have been impacted by SB in this region (Andrus et al., 2020).

2.2. Landsat dataset

All Landsat ARD Tier 1 surface reflectance products between 2001 and 2019 (19 years) with Tile IDs '11/8', '11/9', '10/10' and '10/11' were downloaded for time series analysis. The Landsat ARD products gridded all available Landsat-4, and – 5 TM, Landsat-7 ETM+ and Landsat-8 OLI/TIRS (Thermal Infrared Sensor), and were provided as tiled, georegistered, and atmospherically corrected products defined in a common equal area projection (Dwyer et al., 2018). To speed the data processing, we eliminated the scenes severely contaminated by clouds/shadow, which had <20% clear pixels determined by the quality assessment band. An average of ~1002 ARD scenes across 19 years remained per tile for processing (see Table S1 in the supplementary material).

2.3. Calibration and evaluation plots

Calibration and evaluation sites were identified independently for each study site. Similar to previous Landsat time series studies (Cohen et al., 2017; Kennedy et al., 2010; Zhu et al., 2020), the calendar year was used as the temporal unit of disturbance detection. The initial disturbance year (i.e., reference disturbance timing) is hereafter defined as the earliest year for an occurrence of a spectral signal associated with beetle infestation in the Landsat time series.

We randomly selected 150 calibration sites (circles in Fig. 2) for PIDS model parameter optimization and index selection within the 'forest' category in the 2001 National Land Cover Database (NLCD) (Yang et al., 2018) map for each beetle disturbance agent. The initial disturbance year for each calibration plot was interpreted by toggling to view Landsat time series for red, near infrared (NIR), short-wave infrared (SWIR1 and SWIR2) bands and locating the earliest structural break occurring among multispectral time series curves. If no break was found, the sample was labeled as 'undisturbed'.

Another reference dataset (triangles in Fig. 2) based on field-collected forest disturbance survey data was used to evaluate the beetle-activity detection accuracy for PIDS and other comparative approaches. For MPB, 118 circular plots with a fixed radius (8 m) within lodgepole-pine-dominated stands were selected by random stratified sampling and surveyed in 2010 (Caldwell et al., 2013). For SB, 134 Engelmann spruce plots (20 × 20 m) were surveyed in 2013, 2016, and 2017 (Andrus et al., 2020; Hart and Veblen, 2015). At each plot, tree diameter at breast height (DBH), tree height, tree species, status (live or dead), and likely cause of mortality was collected. The beetle infestation year was determined by evaluating needle colour and amount of remaining needles and branches as described for MPB (Klutsch et al., 2009) and SB (Andrus et al., 2020). Because the field-collected infestation year varied by many years (1–15 years) for individual trees within a single plot, we first generated the field-estimated disturbance year as the disturbance-year category with the maximum basal area, then generated the reference disturbance year by considering the consensus between field-estimate and interpreted disturbance year from Landsat time series visualization. Sample plots with large conflicts (> ± 1 year difference) between field-based estimation and visual interpretation of Landsat time series were eliminated. The reference protocol and the selection results are detailed in the supplementary material (Section S2). Overall, 115 MPB and 115 SB plots were used for evaluating PIDS-detection map accuracy. Considering that the field plots were dominated by MPB and SB infestation, 70 additional plots were randomly sampled within the region without evidence of beetle damage based on the Aerial Detection Surveys (ADS) dataset. While these undisturbed sites were not verified in situ, it was straightforward to confirm them using Google Earth imagery.

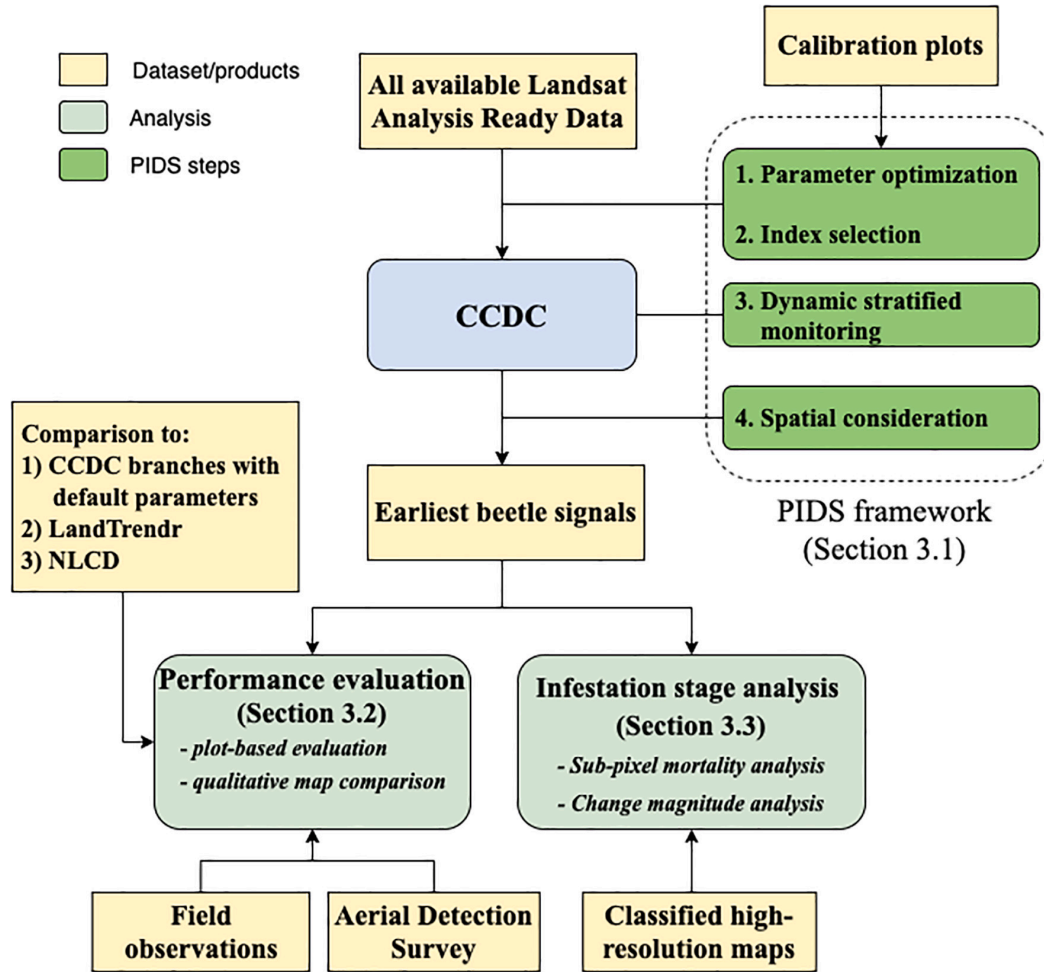


Fig. 3. Flowchart of the dataset used, the methodology and the analysis for this study. CCDC: Continuous Change Detection Classification; PIDS: Parameter optimization, Index selection, Dynamic stratified monitoring, and Spatial consideration; NLCD: National Land Cover Database forest disturbance-date map.

2.4. Aerial detection survey (ADS) data

ADS data were used in a complementary analysis to assess the accuracy of estimated beetle-infestation years at the stand level. The ADS data are polygonal forest health information made by sketch mapping for a variety of specific insects and disease outbreaks, annually reported by US Department of Agriculture (USDA) (Johnson and Wittwer, 2008). The recent ADS data were reported to have 19% omission and 9% commission errors for bark beetle assessment (Coleman et al., 2018). In spite of its relatively coarse minimum mapping unit (5 ha), ADS data are a valuable source for indicating the geographic extent of bark beetle-induced mortality (Hart and Veblen, 2015; Preisler et al., 2012), and the timing of red-stage mortality.

2.5. Classified high-resolution beetle-stage maps

Two classified aerial images (red rectangles in Fig. 2) from previously published studies (Hart and Veblen, 2015; Meddens et al., 2011) were used as reference information for analyzing tree canopy mortality percentage within a Landsat pixel when PIDS detects disturbance for this pixel. An 0.3-m aerial multispectral image was captured on August 2008 covering a proportion of the MPB site, and was classified into five categories (red stage, gray stage, herbaceous, bare soil and shadow) (Meddens et al., 2011). For SB, a classified map (shadows, non-forest,

green trees, and gray stage) was generated from 1-m multispectral aerial imagery captured on August 2011 using a support-vector-machine classifier (Hart and Veblen, 2015). The overall accuracy of the two beetle disturbance maps was ~90% (Hart and Veblen, 2015; Meddens et al., 2011).

3. Methods

As the main method for this study, the PIDS framework was developed to improve the current CCDC procedure. PIDS consisted of four components: 'parameter optimization' and 'index selection' automate CCDC model parameters and index selection using calibration plots as a preprocessing step; 'dynamic stratified monitoring' enables multiple model parameterization to be applied based on thematic information during the CCDC procedure; and 'spatial consideration' incorporates spatial information for post-processing the resultant disturbance maps. We assessed the performance of the PIDS framework for detecting 'the earliest beetle signal' occurring in the Landsat time series, with a comparison to other existing methods and datasets, then linked the earliest signals detected by PIDS to the infestation stages (the workflow is shown in Fig. 3).

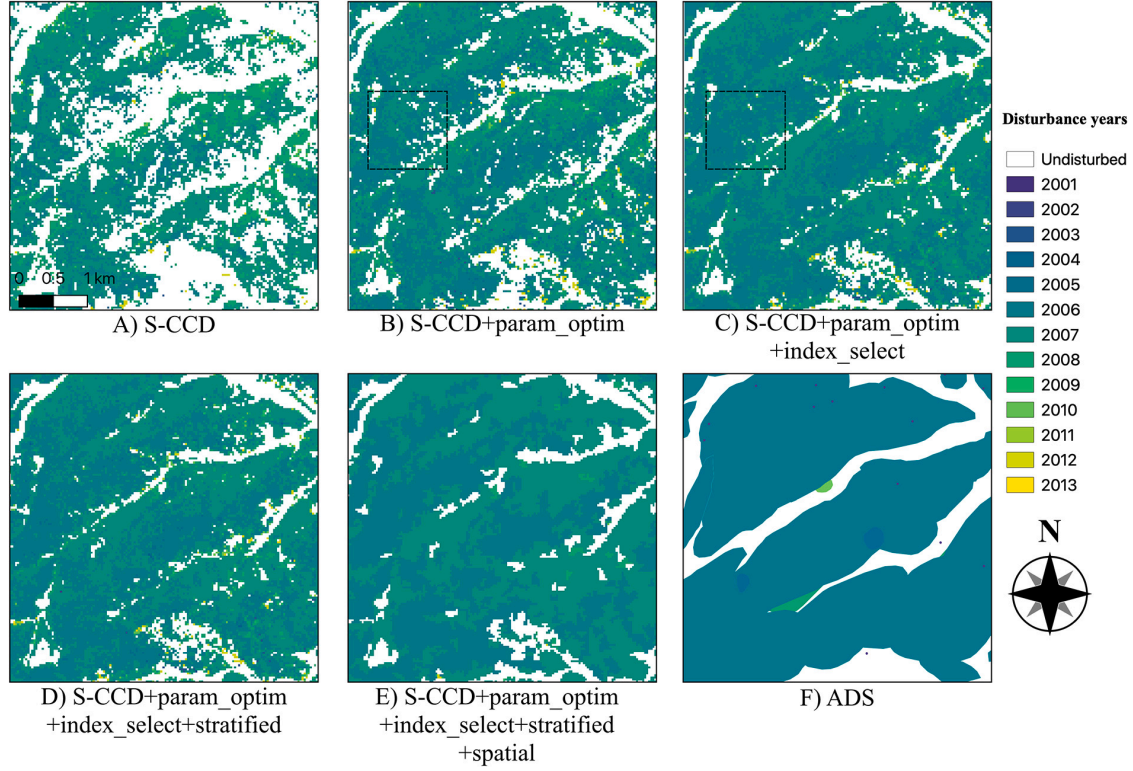


Fig. 4. A subregion (200×200 pixels) disturbed by mountain pine beetle illustrating differences of detection maps among the original Stochastic Continuous Change Detection (S-CCD) algorithm (A), adding in parameter optimization (B, 'param_optim'), index selection (C, 'index_select'), dynamic stratified sampling (D, 'stratified'), spatial consideration (E, 'spatial', namely the PIDS result), and Aerial Detection Surveys data (F, 'ADS') as reference dataset. The black box exemplifies a region showing improvement brought by index selection (the disturbance detection year was advanced for some pixels).

3.1. PIDS framework

3.1.1. Parameter optimization

Parameter optimization focuses on optimizing the most critical parameter for the CCDC-like approach, namely *change probability*. *Change probability* determines a threshold for change magnitudes (CMs). The CM is a single-dimension metric calculated as the minimum norm of standardized change vectors for a number of consecutive observations under scrutiny (called 'peek window'); if a peek window has a change magnitude larger than a critical value of the chi-square distribution $\chi^2_{change_prob}$, the first observation of this peek window is then identified as a breakpoint (Ye et al., 2021; Zhu et al., 2020). The idea of the PIDS approach is to find the optimal threshold to classify CMs between the stable temporal period and the change period extracted from reference samples. Given that N_s and N_c are observation groups, respectively for the stable and the change temporal period, n is a Landsat observation, their representative CMs for these two periods can be defined as:

$$CM_{stable} = \max_{n \in N_s} \{ CM_n * scale \mid Angle(n) < 30^\circ, DistEvi(n) > 0 \} \quad (1)$$

$$CM_{change} = \max_{n \in N_c} \{ CM_n * scale \mid Angle(n) < 30^\circ, DistEvi(n) > 0 \} \quad (2)$$

where CM_n , $Angle(n)$ and $DistEvi(n)$ are respectively the change magnitudes, the change vector angle and the disturbance evidence index for an observation n , and $scale$ is the ratio between the chi-square value based on the defaulted probability ($\chi^2_{Probedef}$) and the chi-square value based on an adjusted probability ($\chi^2_{Probadj}$), which is defined as below:

$$scale = \frac{\chi^2_{Probedef}}{\chi^2_{Probadj}} \quad (3)$$

CM_{stable} are the maximum CMs that represent the CMs closest to occurrence of changes during the stable period; CM_{change} is the highest CM during the change period that holds the most likelihood to be land-cover change. Stochastic Continuous Change Detection (S-CCD) was chosen as the CCDC branch upon which PIDS was built. Compared to other CCDC-like approaches, S-CCD is computationally efficient, more amenable for near-real-time monitoring, and has greater sensitivity to subtle change events (Ye et al., 2021). In S-CCD, the CMs considered as potential breakpoints need to meet two conditions: 1) average angular departure of change vectors to their medium change vector are concentrated (i.e., $Angle(n) < 30^\circ$); 2) the change vector have disturbance evidence (i.e., the sum of the red, negative NIR, and SWIR1 change magnitudes by each Root Mean Square Errors) larger than zero (see Eq. 10 and 11 in (Ye et al., 2021)). The COLD branch has similar restrictions for identifying disturbance-related breakpoints, but slightly different definitions. The variable $scale$ is used to eliminate effects of different temporal density. For a time series denser than the default density, S-CCD and COLD uses the chi-square distribution value ($\chi^2_{Probadj}$) based on an adjusted probability, while others use the chi-square value ($\chi^2_{Probedef}$) based on the defaulted probability (see Eq. S4 in (Zhu et al., 2020)). Instead of adjusting the threshold into a lower value, we used a reverse approach to scale change magnitudes so that magnitudes can be standardized into the default temporal density. If a time-series sample is labeled as an undisturbed sample, then only CM_{stable} was generated for this sample.

We extracted CM_{stable} and CM_{change} from each calibration sample. The best threshold should yield the optimal boundary for separating these

Table 1

Vegetation indices (VIs) selected to detect mountain pine beetle (MPB) and spruce beetle (SB) beetle infestation. NIR: near infrared; SWIR: short-wave infrared.

Index (Abbrev)	Formula	Indicated change	Beetle	Stage of interest	Reference
Red-green Index (RGI)	$RGI = \frac{RED}{GREEN}$	Coloration	MPB	Red stage	Coops et al. (2006); Hicke and Logan (2009); Meddens et al. (2013); Hart and Veblen (2015); Goodwin et al. (2008); Meddens et al. (2013)
Normalized Difference Moisture Index (NDMI)	$NDMI = \frac{NIR - SWIR1}{NIR + SWIR2}$	Moisture stress	SB	Gray stage	
Normalized Burn Ratio (NBR)	$NBR = \frac{NIR - SWIR2}{NIR + SWIR2}$	Moisture stress & needle structure	MPB	Red stage	Meigs et al. (2011); Liang et al. (2014); Senf et al. (2015); Meddens et al. (2013); Hais et al. (2016)
Tasseled Cap Transformation Greenness (TCG)	$TCG = -0.2848 * BLUE - 0.2435 * GREEN - 0.5436 * RED + 0.7243 * NIR + 0.0840 * SWIR1 - 0.1800 * SWIR2$ (Landsat 5/7) OR $TCG = -0.2941 * BLUE - 0.243 * GREEN - 0.5424 * RED + 0.7276 * NIR + 0.0713 * SWIR1 - 0.1608 * SWIR2$ (Landsat 8)	Needle structure	MPB	Red stage	
Tasseled Cap Transformation Wetness (TCW)	$TCW = 0.1509 * BLUE + 0.1793 * GREEN + 0.3299 * RED + 0.3406 * NIR - 0.7112 * SWIR1 - 0.4572 * SWIR2$ (Landsat 5/7) OR $TCW = 0.1511 * BLUE + 0.1793 * GREEN + 0.3283 * RED + 0.3407 * NIR - 0.7117 * SWIR1 - 0.4559 * SWIR2$ (Landsat 8)	Moisture stress	MPB	Red stage	Wulder et al. (2006); Coops et al. (2009); Skakun et al. (2003); Meddens et al. (2013);
Disturbance Index (DI)	$DI = TCB - TCW - TWG$ (TCB: Tasseled Cap Transformation Brightness)	Moisture stress & needle structure	SB	Gray Stage	Hart and Veblen (2015); DeRose et al. (2011)
Normalized Difference Vegetation Index (NDVI)	$NDVI = \frac{NIR - RED}{NIR + RED}$	Needle structure	MPB	Red and Gray Stage	Meddens et al. (2011); Meddens et al. (2013); Assal et al. (2014); Coops et al. (2006); Hart and Veblen (2015);
			SB	Gray stage	

disturbed and undisturbed plots in decision space. Considering that the number of calibration samples is relatively low ($n = 150$), we chose a more generalized modeling approach to obtain the optimal threshold, instead of comparing accuracies for every possible threshold (e.g., sensitivity analysis). a balanced logistic regression (Pedregosa et al., 2011) is applied here:

$$Y = \frac{1}{1 + e^{-(a+b*CM)}} \quad (4)$$

Where Y is the logistic-regression probability for this sample being labeled as disturbance, a and b are the model parameters. The optimal change-magnitude threshold is chosen at $Y = 0.5$, namely

$$CM_{optimal} = -\frac{a}{b} \quad (5)$$

The optimal change probability ($Prob_{change}$) can be computed based on an inverse-chi-squared distribution:

$$Prob_{change} = Inv - \chi^2(CM_{optimal}, k) \quad (6)$$

where k is the number of input variables. A detailed example of logistic regression for computing the change magnitude threshold is presented in Section S3 of the supplementary material.

The above step outputted a change probability threshold (e.g., 0.56 for MPB) lower than the default S-CCD change magnitude (0.95), which can bring a substantial improvement by identifying those disturbed pixels that were missed by the original S-CCD (see an example of MPB disturbance mapping in Fig. 4A to 4B). Note that while multiple inputted bands were used, the overall CMs in CCDC combine univariate magnitudes over the input bands into one measurement, and hence the algorithm is still a single-index problem.

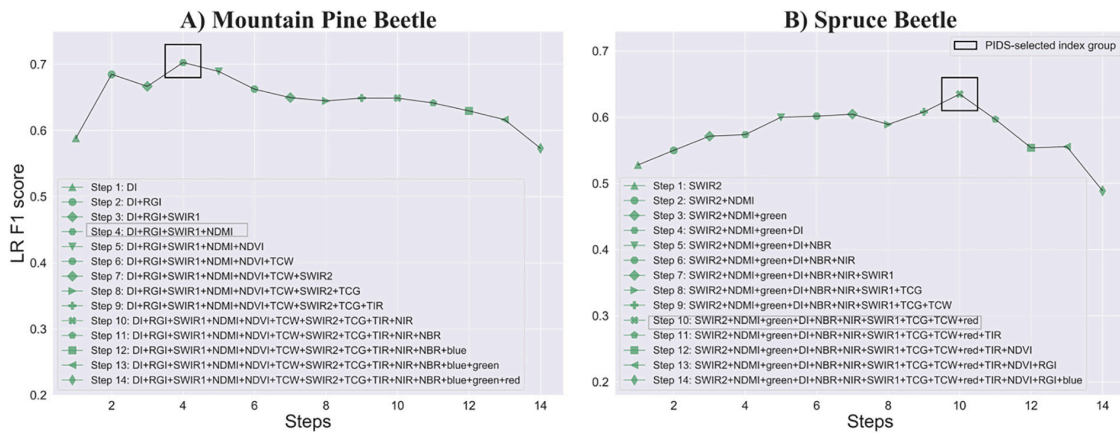


Fig. 5. Stepwise results of forward index selection for mountain pine beetle (MPB, Fig. 6A) and spruce beetle (SB, Fig. 6B) based on logistic-regression (LR) score. The black box highlights the steps (MPB: Step 4; SB: Step 10) that were selected as the final spectral inputs by PIDS. (RGI: Red-green Index; NDMI: Normalized Difference Moisture Index; NBR: Normalized Burn Ratio; TCG: Tasseled Cap Transformation Greenness; TCW: Tasseled Cap Transformation Wetness; DI: Disturbance Index; NDVI: Normalized Difference Vegetation Index; TIR: Thermal infrared). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2

PIDS-based model specifications for mountain pine beetle (MPB) and spruce beetle (SB) (RGI: Red-green Index; NDMI: Normalized Difference Moisture Index; NBR: Normalized Burn Ratio; TCG: Tasseled Cap Transformation Greenness; TCW: Tasseled Cap Transformation Wetness; DI: Disturbance Index).

	Selected index group as model inputs	Prob _{change}
MPB	RGI, NDMI, DI, SWIR1	0.778
SB	NDMI, NBR, TCG, TCW, DI, green, red, NIR, SWIR1, SWIR2	0.907

3.1.2. Index selection

A number of VIs were considered (Table 1): 1) Red-green Index (RGI); 2) Normalized Difference Moisture Index (NDMI); 3) Normalized Burn Ratio (NBR); 4) Tasseled Cap Transformation Greenness (TCG); 5) Tasseled Cap Transformation Wetness (TCW); 6) Disturbance Index (DI); and 7) Normalized Difference Vegetation Index (NDVI). These indices categorize aspects of vegetation changes related to (1) moisture stress, (2) coloration, and (3) needle structure. A forward-selection approach that considers correlation among spectral inputs was developed to select the optimal spectral inputs from a pool of 14 candidate variables (Table 1), including the seven original Landsat bands and seven representative VIs. Forward selection, which has been widely applied in the model selection for stepwise regression (Hair et al., 1998), starts with an empty model, and iteratively adds the band/index to the model that most improves model performance from the previous step. The performance for each selection was evaluated based on the F1 score (Eq. 4) for logistic regression (LR) classification (hereafter called 'LR score').

The LR score indicates the separability between CM_{stable} and CM_{change} . The forward selection keeps adding in the 'best' index sequentially, until all indices have been considered. The index group that first achieves 99% of the highest score during the selection process is viewed as having reached a saturation of score improvement and selected as the optimal spectral inputs for a beetle-specific model. The index selection step can further reduce omission errors and salt-and-pepper noise in the output disturbance map (Fig. 4C). Also, the disturbance year was advanced for some pixels (e.g., dark-green pixels in Fig. 4C versus light-green pixels in Fig. 4B within the black rectangle).

The index-selection results (Fig. 5) show that the LR score for MPB disturbance exhibited a noticeable rise (from 0.588 to 0.685) at Step 2 when the RGI was selected and added into the model, reaching the highest score (0.703) at Step 4 when NDMI was selected in the model, and then gradually decreased when additional indices were added (Fig. 5A). The LR score for SB increased until Step 10 where it peaked (0.635), then substantially declined (Fig. 5B). The final PIDS model specifications are provided in Table 2. Step 4 and Step 10 were chosen as the final spectral inputs, respectively for MPB and SB.

3.1.3. Dynamic stratified monitoring

The model parameters and selected indices are determined for a specific subtle change event, and therefore might be overly sensitive when applied to land cover types where that specific type of subtle change does not occur. Therefore, a dynamic procedure was developed that allows for automatic updating of thematic information once a breakpoint is detected and chooses a model specification contingent on the updated land cover type. A seven-category land cover system was used (see Section S4 in the supplementary material), mostly following the definition of primary land-cover types of LCMAP map products (Brown et al., 2020), except we excluded 'ice/snow', which had limited extent in our study sites (< 0.01%). The 2001 NLCD land-cover map (Yang et al., 2018) was used as the initial base map, which was aligned with the initial year of the Landsat time series (i.e., 2001).

CCDC-like algorithms have an initialization stage to define a temporal window for a 'stable' land condition that is characterized by a group of LASSO coefficients (Tibshirani, 1996). The LASSO coefficients were used to generate the predicted spectral reflectance on July 1 of the first year (Brown et al., 2020) of the initialization stage, forming

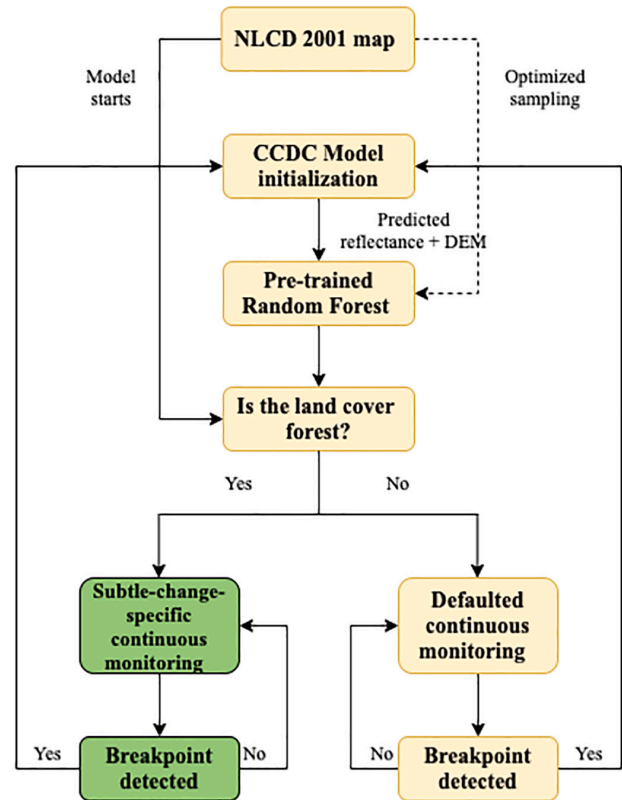


Fig. 6. Flowchart of dynamic stratified monitoring. CCDC: Continuous Change Detection Classification, NLCD: National Land Cover Database.

predictor variables corresponding to six multispectral Landsat bands (TIR band was excluded). The predicted spectral reflectance was updated once a new initialization stage was finished following a detection of a break. Additionally, we added a static variable representing elevation as a seventh predictor, because Zhu et al. (2016) reported that the elevation was the most influential variable to CCDC-based forest classification.

A pre-trained classification tree model (Brown et al., 2020; Chen and Guestrin, 2016) was built offline using the seven predictors (6 spectral bands + elevation) from the first initialization stage and land-cover types as the response variable, based on a set of optimized calibration samples from the procedure of Zhu et al. (2016) (Section S4 in the supplementary material). Ten-fold cross validation indicated a satisfactory overall accuracy for the resultant pre-trained model (81.9% for the MPB site; 82.4% for the SB site) and a high categorical accuracy for forest (omission errors were 8.5% and 8.4%, and commission were 13.8% and 11.0%, respectively for the MPB and SB sites, see Section S4 in the supplementary material). Once a new initialization was completed, the land-cover type for the ongoing monitoring period was updated by inputting the predicted references derived from the new initialization stage into the pre-trained classification tree model. If the predicted land-cover type was forest, the beetle-specific model was applied for the follow-up monitoring until the next breakpoint; if the prediction was one of other land types, the default model was used.

3.1.4. Spatial consideration

Spatial context was incorporated during post-processing by assuming that land-cover change often occurs in a homogeneous patch. A two-fold four-connected sieving filter was applied to eliminate spatially isolated pixel clumps in the disturbance-year maps (Fig. 4E). A Minimum Mapping Unit (MMU) of 0.5 ha was used as the sieving threshold (Senf et al.,

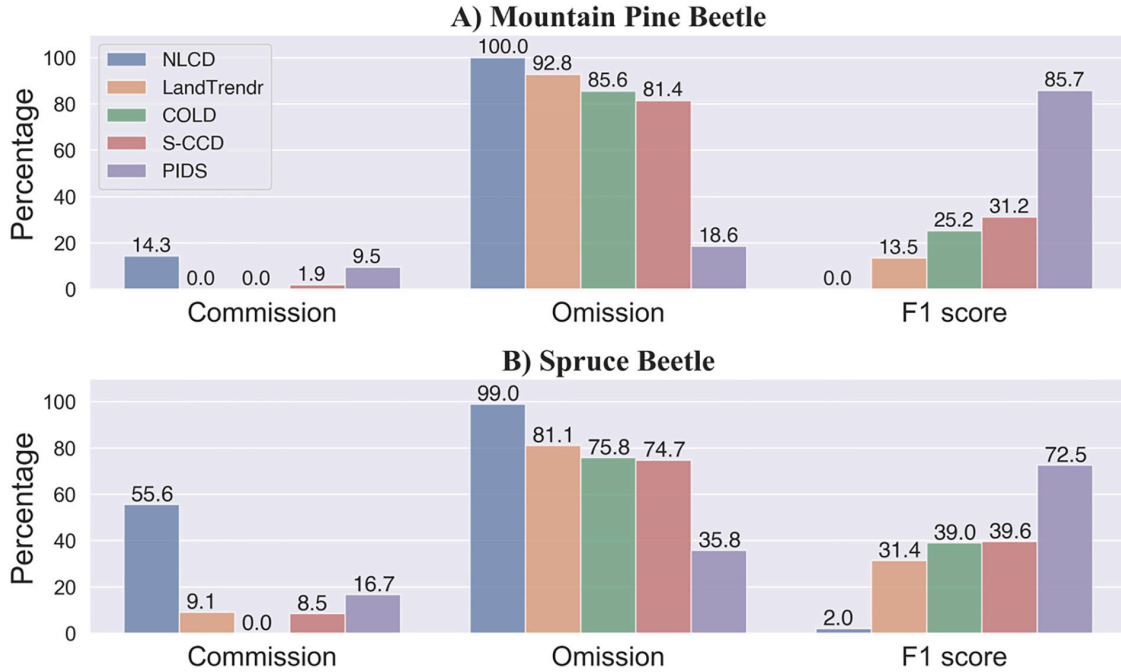


Fig. 7. Performance evaluation results for NLCD forest disturbance-date products, LandTrendr, COLD (change probability = 0.95), S-CCD (change probability = 0.95), and PIDS against reference plots for mountain pine beetle (A) and spruce beetle (B) (NLCD: National Land Cover Database; COLD: Continuous monitoring of Land Disturbance algorithm; S-CCD: Stochastic Continuous Change Detection algorithm).

2017), representing the minimum disturbed/undisturbed forest patch size (roughly seven Landsat pixels). The first sieving filter was applied to a binary ‘disturbed vs. undisturbed’ map, which was then reclassified from the disturbance-year map generated by previous steps, with the goal to remove isolated false positive pixels and restore small patches of omission errors. The second sieving filter was applied to the disturbance-year maps using the first-sieving result as a mask, with the goal of homogenizing the disturbance years over space and assigning disturbance years to those ‘small holes’ of omissions that were filled in the first step. The difference compared to the original sieving filter was that two-folded filter was designed to retain those infestation disturbance clumps whose spatial neighbors were dominated by a broad category of ‘infestation’ but the area of undisturbed category was larger than that of any single disturbance-year category.

3.2. Performance evaluation for PIDS

The large spatial extent and long time period of our study posed challenges for obtaining a large number of high-quality reference observations ideally suited for our purpose. To evaluate the results, we included a comparison with field observations from previously published studies, aerial surveys of bark beetle damage, and results from other remote sensing algorithms/products. We compared the PIDS results to the two CCDC branches, COLD (Zhu et al., 2020) and S-CCD (Ye et al., 2021), another commonly used algorithm based on annual Landsat time series, LandTrendr (Kennedy et al., 2010), and the NLCD forest disturbance date products (note: this is different than the land-cover product used in the mapping algorithm) (Yang et al., 2018). Breakpoints identified within fire perimeter polygons within each fire-active temporal window were eliminated for all products. COLD and S-CCD used 0.95 as the change-magnitude threshold because this threshold achieves the best F1 score for forest disturbance detection from the previous study (Ye et al., 2021). We selected LandTrendr because it is reportedly more sensitive to subtle changes compared to other forest change detection algorithms (Cohen et al., 2017). We used

the Google Earth Engine implementation of LandTrendr with the optimal parameters provided by Kennedy et al. (2018). Rodman et al. (2021) reported that the accuracy of LandTrendr for beetle disturbance detection was insensitive to parameter selection (see their Appendix A). The NLCD forest disturbance products (available from <https://www.mrlc.gov/nlcd-2016-science-research-products>) were generated from the combination of information from the NLCD 2016 change detection, land cover classification, and the LANDFIRE Vegetation Change Tracker (VCT) disturbance products.

Performance evaluation for disturbance years detected by PIDS and other methods include 1) plot-based assessment; 2) qualitative map comparison. Plot-based accuracies based on reference plots were reported using *omission*, *commission* and *F1 score* (Zhu et al., 2020). True positives (*TP*) were defined as mapped disturbance years equal to or one year earlier than the reference year; true negatives (*TN*) were defined as correctly mapped undisturbed sites. The *omission*, *commission* and *F1 score* were defined as:

$$omission = \left(1 - \frac{TP}{Ref_{presence}}\right) * 100\% \quad (7)$$

$$commission = \left(1 - \frac{TN}{Pred_{presence}}\right) * 100\% \quad (8)$$

$$F1\ score = 2 * \frac{(1 - commission) * (1 - omission)}{2 - commission - omission} \quad (9)$$

Where $Ref_{presence}$ and $Pred_{presence}$ were the total sample number for reference beetle presence and predicted beetle presence respectively. In addition, the quantitative PIDS evaluation was further determined through different error types. Five error types were reported with their percentages in our evaluation samples: 1) complete miss; 2) one year earlier; 3) multiple years earlier; 4) one year later; and 5) multiple years later than the reference disturbance years. The disturbance-year maps produced by PIDS and other methods were also qualitatively compared

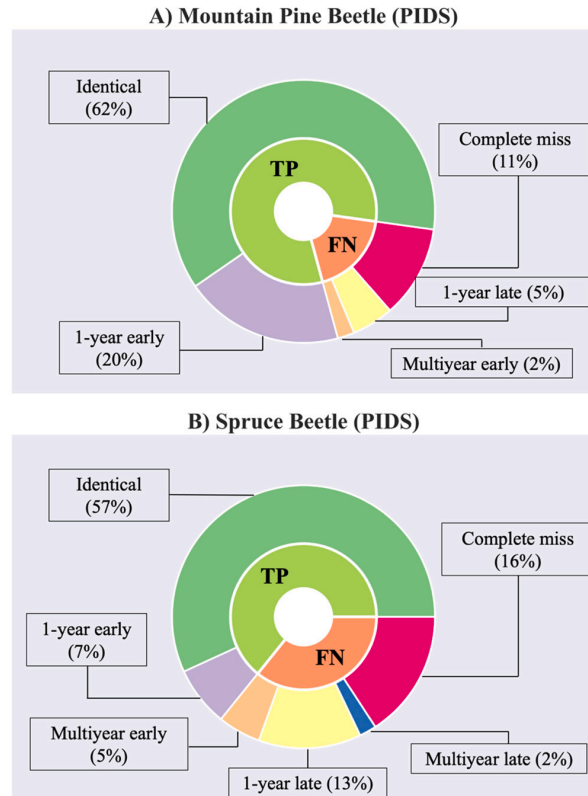


Fig. 8. Percentages of different types for true positives (TP) and false negative (FN, omission errors) of the PIDS results based on reference plots. (A) mountain pine beetle. (B) spruce beetle ('TP' - 'true positive' defined for this study, i.e., identical or 1-year early; 'FN' - 'False negative', namely omission errors).

and assessed at the stand/regional scale through comparison with ADS data.

3.3. Infestation stage analysis

We spatially aggregated the high-resolution classified aerial imagery to create mortality percentage maps (i.e., the percentage of dead canopy cover within a grid cell) at 30-m resolution and aligned them to the 30-m PIDS-detected disturbance-year map (note that the SB site only has the gray-stage mortality percentage map while the MPB has both red- and gray-stage mortality percentage maps). Sub-pixel infestation-stage percentages were calculated for the pixels whose disturbance years labeled by PIDS coincide with the aerial image acquisition years. Given that the single-date high-resolution images included the pixels at different infestation stages, we applied a 'space-for-time substitution' strategy (Blois et al., 2013) that uses spatial phenomena to predict temporal change of the sub-pixel mortality. The average infestation-stage percentages were calculated for the Landsat pixels grouped by their PIDS-detected years, representing sub-pixel mortality conditions before versus after detection. The infestation-stage percentages were further compared to the empirical thresholds for confirming beetle infestation in the previous studies, i.e., 50% red-stage mortality for MPB (Meddens et al., 2013) and 40% gray-stage mortality for SB (Hart and Veblen, 2015).

The change magnitudes (CMs), which can reveal spectral attributes of the breakpoints found in the time series, were used as a complementary approach to confirm the infestation stages detected by PIDS. CMs were reported for the selected indices for each beetle model

specification. CMs for the five original Landsat bands were also reported to illustrate a more complete beetle-induced spectral change for stage analysis, though some of them were not included in the beetle-specific models.

4. Results

4.1. Performance evaluation for PIDS

4.1.1. Plot-based evaluation

The quantitative evaluation based on reference plots (Fig. 7) illustrated that PIDS achieved the best overall performance among the different approaches; the resulting F1 score was higher than those from other datasets/approaches by at least 30% F1 score improvement. The improvement in F1 score was due to the significantly reduced omission error of PIDS compared with other products. The NLCD forest disturbance product failed to capture all positive cases (omission $\approx$ 100%) for both MPB and SB. Similarly, LandTrendr, and the original S-CCD and COLD resulted in high omission error rates above 70% for both beetle disturbance scenarios. S-CCD and COLD have similar F1 scores (MPB: 0.312 vs. 0.252; SB: 0.396 vs. 0.390,) but were greater than LandTrendr F1 scores (MPB: 0.135; SB: 0.314). Note that the evaluation metrics reported here were used only for comparing the relative performance differences and might not reflect the 'real' detection accuracy meaning that the number of correct predictions by the total number of pixels, because our evaluation samples were collected from a non-probabilistic sampling procedures.

The PIDS evaluation was determined through different error types (Fig. 8). A strict definition for true positive (TP) (either 'identical' or 'one year early') was used that resulted in omission errors associated with correct detections but incorrect disturbance years. If we include 'one year later than reference' as a true positive as has been done in other studies (Cohen et al., 2017; Coops et al., 2020), the omission error rate decreased from 19% to 13% for MPB and from 36% to 23% for SB. Compared with MPB, the generally lower detection rate for SB was likely caused by the mixed nature of host stands and an absence of an obvious red stage.

4.1.2. Qualitative map comparison

A comparison of the MPB disturbance-year maps generated by PIDS, other approaches/datasets, and the ADS maps is shown in Fig. 9. The maps produced by COLD are similar to that of S-CCD, so only the maps of S-CCD are presented here. Across each study site, the disturbance-year map of PIDS (Fig. 9C) was more similar to ADS (Fig. 9D) than results from LandTrendr (Fig. 9A), and the original S-CCD (Fig. 9B). Fig. 9 E-H shows an enlarged view for an example forest stand. The LandTrendr missed most of the beetle disturbance in this stand (see Fig. 9E). The original S-CCD missed a small number of disturbed patches and showed a delay by 1–3 years in disturbance detection, compared with PIDS and ADS (denoted as darker green). PIDS (Fig. 9G) showed a similar spatial coverage and disturbance timing pattern as ADS (Fig. 9H). Further examination based on a high spatial-resolution imagery (Fig. 9I) illustrated that PIDS (Fig. 9G) derived a finer-grained spatial pattern reflecting the presence of roads/trails and small undisturbed bare lands, whereas ADS represented a coarser delineation for the landscape given its 5-ha minimum mapping unit (Fig. 9H).

The beetle-infestation region mapped by PIDS (Fig. 10C) had good agreement with the ADS map, whereas LandTrendr (Fig. 10A) missed most of the SB infestation area and S-CCD (Fig. 10B) missed some of them. This finding agreed well with the plot-based evaluation. Discrepancies between mapped disturbance years occurred between PIDS and ADS in the center and the southeast regions. Fig. 10E-H exemplifies a region that ADS showed as having an earlier beetle disturbance in 2005 (denoted as dark blue) whereas PIDS indicated that the timing of infestation was between 2009 and 2011. This discrepancy may be due to data uncertainties in the earliest ADS data collections (Johnson and

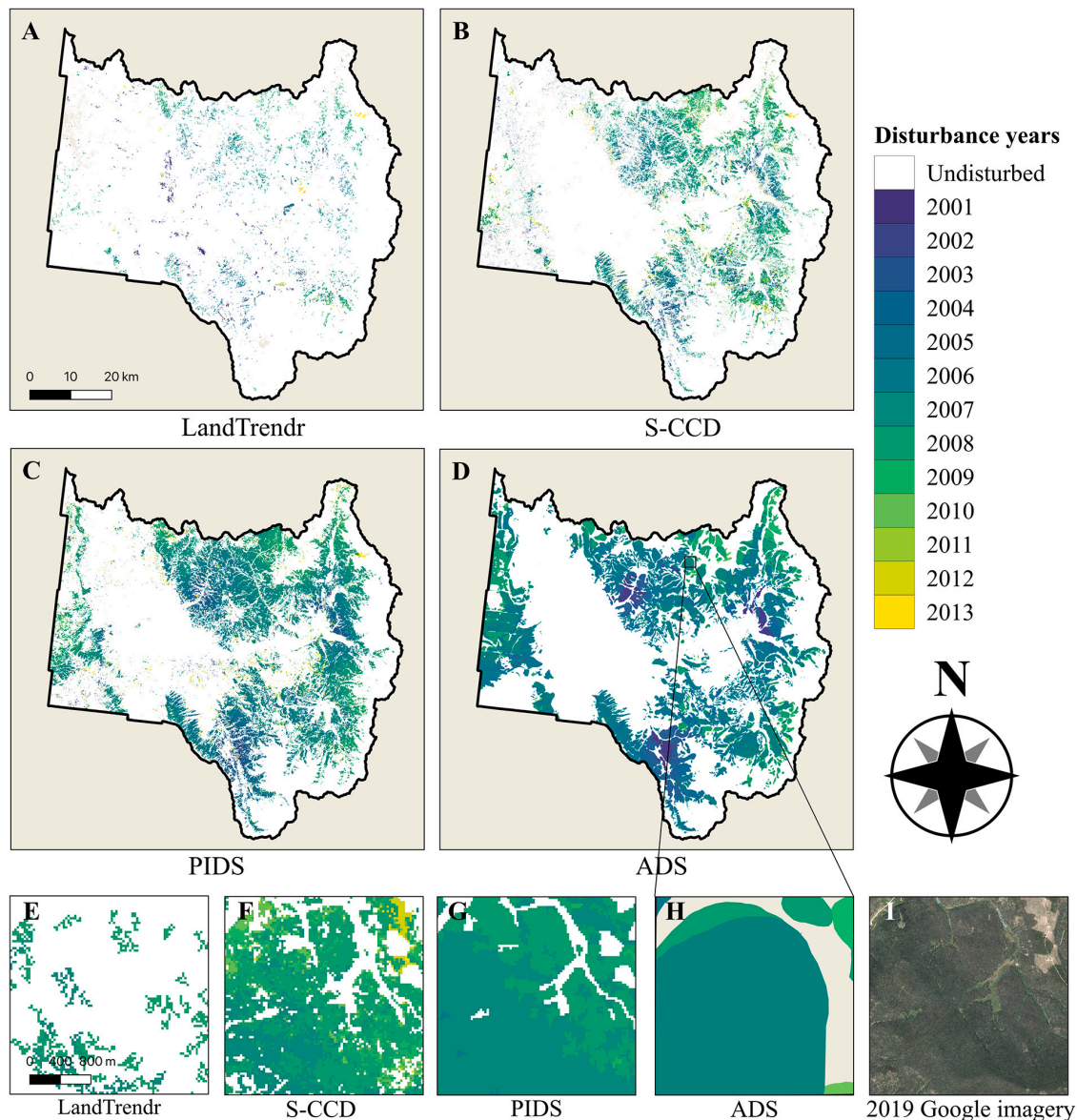


Fig. 9. Comparison of LandTrendr with default parameters (A), S-CCD with the default threshold of 0.95 (B), PIDS (C) and ADS (D) for mapped disturbance years for the mountain pine beetle site. The colour denotes disturbance years: the spectrum ranging from blue to yellow corresponds to the years from 2001 to 2013. (E) – (I) exemplify stand-level differences of different approaches for a subregion with a high-resolution image as reference: PIDS shows an earlier detection than S-CCD, and a finer spatial pattern compared with the ADS data. S-CCD: Stochastic Continuous Change Detection algorithm; ADS: Aerial Detection Surveys data. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Wittwer, 2008) or errors in PIDS. Generally, PIDS is more spatially precise, whereas ADS polygons covered non-forest regions such as bare land and ponds in the aerial imagery.

4.2. Infestation stage analysis

The results of coupling classified high-resolution imagery and PIDS disturbance-year maps reveal that when the disturbances were first detected by PIDS ('0 year' in Fig. 11), the average cumulative red-stage mortality percentage (hereafter referred to the sum of red- and gray-stage cover percentage) was 55.9% for MPB, and the gray-stage mortality percentage averaged 55.3% for SB. The mean mortality percentages was calculated for those Landsat pixels where high-resolution imagery acquisition was captured one, two years prior to PIDS-detected

years (i.e., '-1 year' and '-2 year' in Fig. 11) and the other pixels for one- and two-year old detections based on PIDS-detected years (i.e., '+1 year' and '+2 year' in Fig. 11). The mortality percentage for the pixels prior to the detection (i.e., '-1 year' and '-2 year' in Fig. 11) were all lower than the empirical thresholds (dashed lines) that respectively defined pixel-level red stage for MPB and gray stage for SB (Hart and Veblen, 2015; Meddens et al., 2013), indicating that the regions that had not yet been detected by PIDS were generally still 'undisturbed pixels'. In the other words, PIDS started detecting beetle infestation during the first year that visual attacking signals (i.e., red and gray stages) were shown at a Landsat-pixel level. For MPB, one-year old infestation ('+1 year' in Fig. 11) pixels had a higher red-stage percentage than that of '0 year', which implies that the red foliage was still increasing after the disturbance was detected. This finding also indicates that PIDS captures the

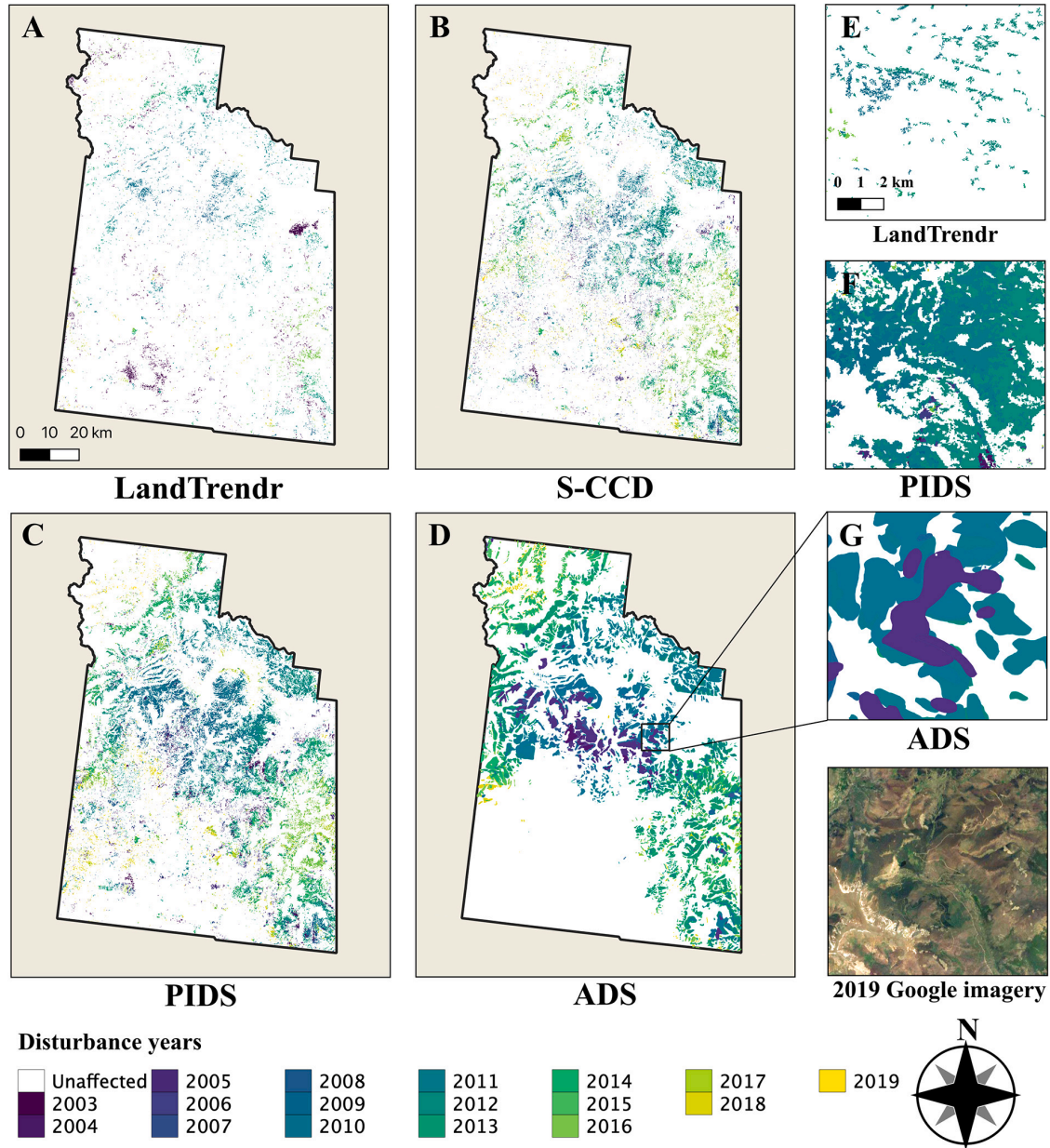


Fig. 10. Comparison of LandTrendr (A), S-CCD (B), PIDS (C) and ADS (D) to indicate mapped disturbance years for the spruce beetle site between 2003 and 2019. (E) – (H) shows stand-level differences for a subregion: ADS (G) suggests that the first infestation year was 2005, much earlier than PIDS (F) which was around 2010, which is possibly due to commission errors in early ADS data products or PIDS errors; PIDS portrays a finer spatial pattern for beetle damage progression (i.e., an eastward spreading trend that is not portrayed in the ADS map). The LandTrendr (E) misses detection for most of beetle-disturbed regions. S-CCD: Stochastic Continuous Change Detection algorithm; ADS: Aerial Detection Surveys data.

change signal at the initial red stage.

Change magnitudes for moisture-stress-related indices (e.g., DI, SWIR2, NDMI, TCW) were noticeably higher than pigment-related indices (e.g., RGI, Red and Green), as well as needle structure-related indices (e.g., TCG and NIR) (Fig. 12). This spectral pattern indicates that infestation symptoms were dominated by increased water stress, rather than discoloration (characteristics of red stage) or needle drop (characteristics of gray stage), when the breakpoints were detected. This result confirms the finding from the above mortality percentage analysis, i.e., PIDS captured change signals at the initial red stage for MPB, and the initial gray stage for SB.

5. Discussion

The results of this study show that choice of VIs was important for subtle change detection: several VIs that are sensitive to early onset of water stress in the canopy associated with beetle infestation, such as DI and NDMI, were favored by the index selection procedure as model inputs for both beetle disturbance agents, and also reached relatively high rankings in the single-index performance evaluation (Table S5 in the supplementary material) compared to the Landsat spectral bands. This finding differs from previous CCDC studies that suggested that VIs did not bring an obvious performance improvement for a generic land

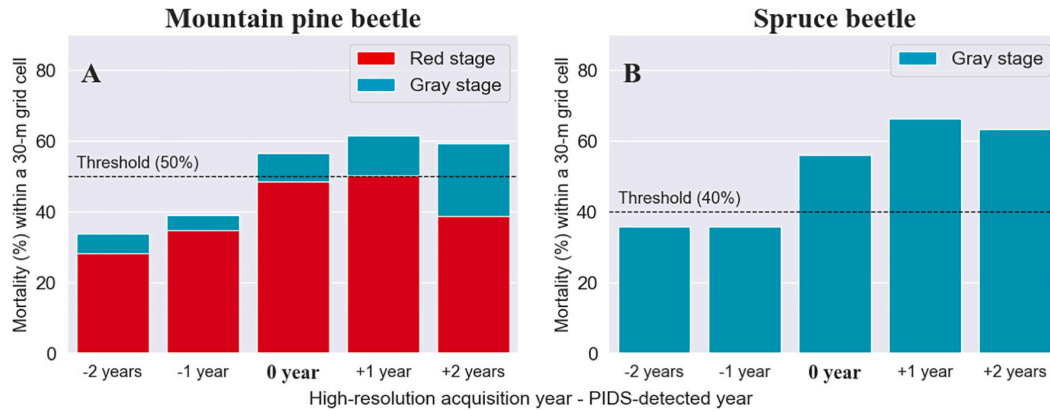


Fig. 11. Average percent mortality of high-resolution reference maps aggregated to 30-m spatial resolution plotted by first year of disturbance estimated by PIDS, relative to the acquisition year of high-resolution maps. “0” means that the PIDS detection year matched the high-resolution acquisition year, “-1” means that the PIDS detection year was earlier, and so on. (A) mountain pine beetle site, partitioned into red versus gray stage using reference map. (B) spruce beetle site. Thresholds for defining ‘undisturbed versus disturbed’ for mixed pixels, used in previous studies, are shown as dashed horizontal lines. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

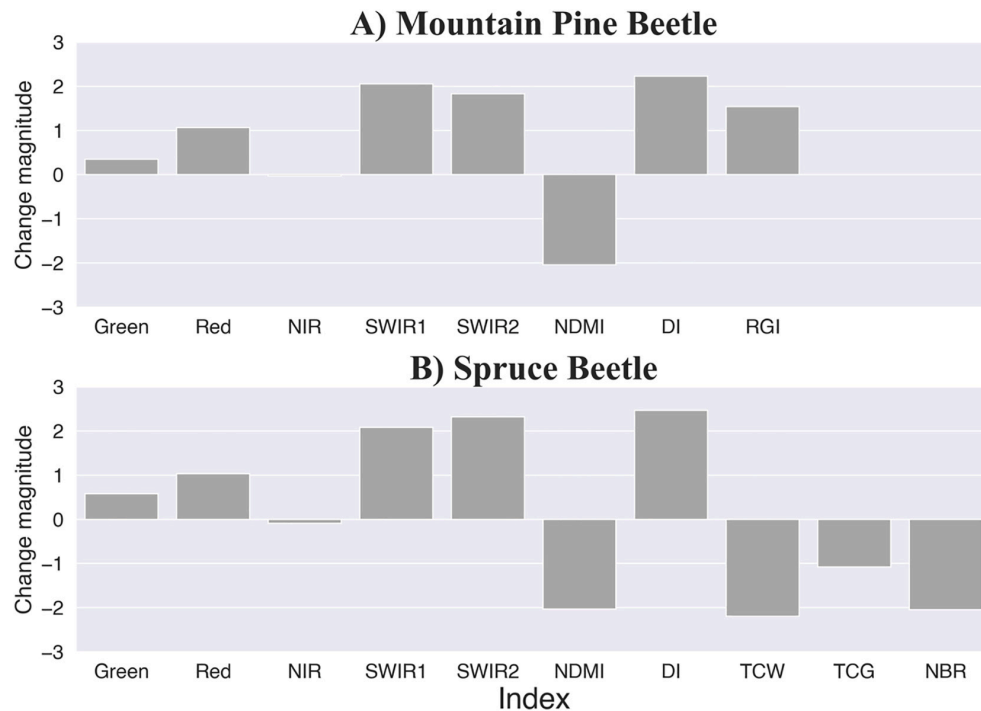


Fig. 12. Average change magnitudes associated with PIDS-detected breakpoints for the mountain pine beetle site (A) and the spruce beetle site (B). Five spectral bands and the indices selected in each beetle model specification were included. Moisture-stress-related indices such as SWIR2, NDMI and TCW had higher absolute change magnitudes than the coloration-related indices such as Red and needle structure-related indices such as TCG and NIR, which indicates that the spectral change signals captured by PIDS are dominated by moisture stress, rather than coloration change or needle structure change. RGI: Red-green Index; NDMI: Normalized Difference Moisture Index; NBR: Normalized Burn Ratio; TCG: Tasseled Cap Transformation Greenness; TCW: Tasseled Cap Transformation Wetness; DI: Disturbance Index. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

disturbance monitoring (Zhu et al., 2020), but agrees with other literature related to remote sensing of beetle mortality (Table 1). DI ranked first for MPB (i.e., first picked by index selection), while SWIR2 was the best for SB, which agrees with the finding of Foster et al. (2017) that reported Landsat SWIR2 as the most useful wavelength range (2110–2290 nm) for early-stage spruce beetle infestation based on field spectroradiometer data.

There were some differences in the selected indices between MPB

and SB: RGI was selected into the MPB-specific model, but was not chosen for the final index group for SB. This coincides with the findings of Hart and Veblen (2015), where RGI was not identified as a useful indicator of spruce beetle infestation because SB infestations do not exhibit an obvious red stage (Hart and Veblen, 2015). The SB model additionally included NIR and NBR indices that are indicative of forest structure, because needles of infested Engelmann spruce begin dropping when change signals were detected by PIDS. The SB model required

many more inputted indices than the MPB model (10 indices versus 4 indices), possibly because SB disturbances require a larger ensemble of spectral inputs to capture its subtler spectral signals. These differences justify the necessity of adapting detection models separately for different change agents due to their distinct biological attributes.

The PIDS results were compared to original CCDC-like approaches, disturbance mapping products and aerial survey data. PIDS provides a supervised solution that optimizes model parameters and inputs of CCDC-like algorithms based on plot-based disturbance samples, therefore markedly enhancing the detection performance compared with original CCDC-like algorithms, which are designed for a generic change mapping. Another common approach based on annual time series, LandTrendr, presents relatively high omission error rates for both beetle species; similarly, a low detection rate (26.5%) was reported by Rodman et al. (2021) who applied LandTrendr for mapping spruce beetle disturbance in Colorado. This can be explained by the difficulty of separating subtle change signals from other sources of spectral variations, such as variation in vegetation phenology, if only representative annual observations were chosen for constructing time series (Rodman et al., 2021). Our work is closely related to the recent work done by Bullock et al. (2020b) who modified the CCDC algorithm to detect tropical forest degradation. In their work, a spectral index called the 'Normalized Degradation Fraction Index' was chosen based on the related previous work in the tropical region (Bullock et al., 2020b), while our study focused on developing generic framework that can be used to automatically select spectral indices and tune model parameters based on in situ data.

Because a 30-m Landsat pixel can be a spatial mixture of undisturbed trees, trees at different infestation stages, and other undisturbed land-cover types, we reported sub-pixel percentage of characteristic stages at the year that beetle signals were captured by PIDS. The results show that PIDS first identifies mortality when roughly 50% of a Landsat pixel has visual signs of beetle infestation, i.e., a transitioning period between green and red/gray stage. Despite the inability to confidently capture pre-visual stage (green stage) mostly due to spectral radiometric and spatial limitations of Landsat sensors, PIDS can achieve comparable disturbance-detection timeliness to ADS data. Aerial surveying was performed with high data collection costs (\$0.25/ha; (Johnson and Ross, 2008)); their spatial quality is often unsatisfactory because the aircraft is moving at a high speed and surveyors are forced to make quick interpretation decisions (Coleman et al., 2018), as well as a relatively coarse spatial grain (minimum mapping unit as 5 ha), but with the potential to identify low-density subtle changes (e.g., infestation of individual trees). The implementation of PIDS is only contingent on calibration samples which could be generated by visual interpretation of Landsat time series, requiring much less manual work than ADS. The disturbance years mapped by PIDS and ADS were generally comparable at the landscape scale, but PIDS allowed for finer delineation at the stand scale because of the finer spatial resolution of Landsat satellite data. For operational disturbance monitoring, S-CCD, the base algorithm of PIDS, is a near real-time algorithm that allows PIDS to detect forest disturbances as close in time to when they occur as possible (Ye et al., 2021).

When operationalizing PIDS for mapping a large area, one potential concern is that PIDS is unable to differentiate subtle change from multiple change agents, such as stand-replacing disturbances. To address this issue, a two-step hierarchy of change detection could be potentially applied in future. In the first step, PIDS is used to optimize model parameters and spectral inputs in term of 'the minimum change of interest', enabling a detection of the breakpoints for both subtle change and other land conversion. Then, change agent classification can be developed to separate subtle and drastic changes by characterizing multi-spectral change magnitudes associated with these breakpoints. Another problem for subtle-change studies is the collection of an adequate number of evaluation samples for subtle change events (Vogelmann et al., 2016). We recommend the response design for subtle change mapping should be made with the aid of field observations to assure the

accuracy of the interpreted date of disturbance. However, typically few large area historical field inventory records are publicly available for this purpose (Fortier et al., 2011). Given by this limitation, we also chose an extensive map-based qualitative assessment and evaluated the resultant spatial patterns of detected disturbances against other reference data sources such as aerial surveying data.

Lastly, this study aimed to generalize four useful components for detecting a specific type of subtle change from satellite-based time series data with an example implementation for each component. Future work could focus on improving the current implementation for some components. For example, spatial information holds great potential for improving subtle disturbance detection, and has not been fully explored in this study; the future research can be directed to incorporating spatial synchronization of increasing change magnitudes into time-series break decision, other than merely for post-processing.

6. Conclusion

In this study, we developed and tested a four-component framework named 'PIDS' to map subtle changes (early-stage tree mortality caused by bark beetles) using dense Landsat time series. As a supervised time series framework, PIDS improved model sensitivity by incorporating calibration samples, and hence greatly increased overall performance for detecting early signals of beetle disturbances. Accuracies from PIDS were higher than other algorithms that also used Landsat time series as well as other forest disturbance products. Our index-selection results revealed the importance of vegetation indices for beetle-related subtle change detection. Further investigation illustrated that the change signals captured by PIDS correspond to a transitioning period between green and fully red/gray stage to a Landsat pixel. Overall, this study advances the ability of satellite-based time series to accurately map subtle changes such as forest condition change and provides a solution for extracting other similar subtle disturbances (e.g., windblown) from time series by leveraging user-provided samples.

Declaration of Competing Interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.rse.2021.112560>.

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